| | |
|---|---|
| **Status** | Finished |
| **Started** | Sunday, 18 October 2026, 10:16 PM |
| **Completed** | Sunday, 18 October 2026, 11:06 PM |
| **Duration** | 49 mins 59 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

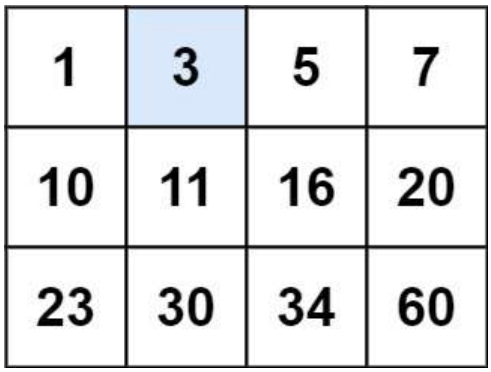
You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

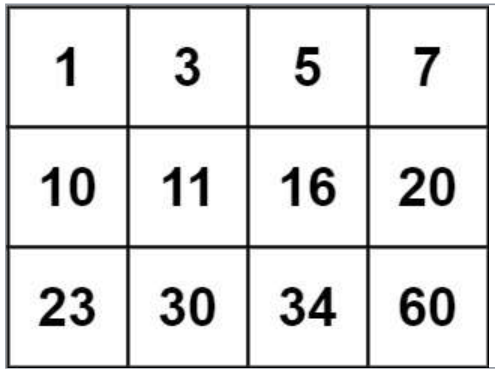
You must write a solution in `O(log(m * n))` time complexity.

**Example 1:**




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

**Example 2:**




```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

**For example:**

| Test | Result |
|---|---|
| print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13)) | False |
| print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3)) | True |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def searchMatrix(matrix, target: int) -> bool:
    a = sum(matrix,[])
    l,r=0,len(a)-1
    while l <= r:
        m = (l+r)//2
```

```
        if a[m] == target:
            return True
        if a[m] < target:
            l = m + 1
        else:
            r = m - 1
    return False
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13)) | False | False | ✓ |
| ✓ | print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3)) | True | True | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

**Example 1:**

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

**Example 2:**

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

**Constraints:**

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

**For example:**

| Test | Result |
|---|---|
| `print(search([-1,0,3,5,9,12],9))` | 4 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
from typing import List
def search(nums: List[int], target: int) -> int:
    l,r=0,len(nums) - 1
    while l<= r:
        m = (l+r)//2
        if nums[m]==target:
            return m
        elif nums[m]<target:
            l=m+1
        else:
            r=m-1
    return -1

```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | `print(search([-1,0,3,5,9,12],9))` | `4` | `4` | ✓ |
| ✓ | `print(search([-1,0,3,5,9,12],2))` | `-1` | `-1` | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

**Input Format**

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

**Output Format**

**Print** peak numbers separated by space.

**Sample Input**

5

8 9 10 2 6

**Sample Output**

```
10 6
```

**For example:**

| Input | Result |
|---|---|
| 4<br>12 3 6 8 | 12 8 |

**Answer:** (penalty regime: 0 %)

```
n = int(input())
a = list(map(int,input().split()))
for i in range(n):
    if i == 0:
        if a[i] >= a[i+1]:
            print(a[i],end=" ")
    elif i == n-1:
        if a[i] >= a[i-1]:
            print(a[i],end=" ")
    else:
        if a[i] >= a[i-1] and a[i] > a[i+1]:
            print(a[i],end=" ")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 7<br>15 7 10 8 9 4 6 | 15 10 9 6 | 15 10 9 6 | ✓ |
| ✓ | 4<br>12 3 6 8 | 12 8 | 12 8 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

**Input Format:**

The first line contains S1.
The second line contains S2.
The third line contains N.

**Output Format:**

The first line contains the N characters present in S1 which are also present in S2.

**Boundary Conditions:**

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

**Example Input/Output 1:**

Input:

abcbde
cdefghbb
3

Output:

bcd

**Note:**

b occurs twice in common but must be printed only once.

**Answer:** (penalty regime: 0 %)

```
s1 = input()
s2 = input()
n = int(input())
ans = ""
for c in s1:
    if c in s2 and c not in ans:
        ans += c
    if len(ans) == n:
        break
print(ans)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | abcbde<br>cdefghbb<br>3 | bcd | bcd | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

**Input Format**

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

**Output Format**

Print Yes or No.

**Sample Input**

7

0 1 2 4 6 5 3

1

**Sample Output**

Yes

**For example:**

| Input | Result |
|---|---|
| 5<br>8 9 12 15 3<br>11 | Yes |
| 6<br>2 9 21 32 43 43 1<br>4 | No |

**Answer:** (penalty regime: 0 %)

```
n = int(input())
a = list(map(int,input().split()))
k = int(input())
for i in range(n):
    for j in range(i+1,n):
        if a[i] + a[j] == k:
            print("Yes")
            break
    else:
        continue
    break
else:
    print("No")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 5<br>8 9 12 15 3<br>11 | Yes | Yes | ✓ |
| ✓ | 6<br>2 9 21 32 43 43 1<br>4 | No | No | ✓ |
| ✓ | 6<br>13 42 31 4 8 9<br>17 | Yes | Yes | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Write a Python program for binary search.

**For example:**

| Input | Result |
|---|---|
| 1,2,3,5,8<br>6 | False |
| 3,5,9,45,42<br>42 | True |

**Answer:** (penalty regime: 0 %)

```
x = list(map(int,input().split(',')))
a = int(input())
x.sort()
l = 0
h = len(x) - 1
while l <= h:
    mid = (l+h)//2

    if x[mid] == a:
        print(True)
        break
    elif x[mid] < a:
        l = mid + 1
    else:
        h = mid - 1
else:
    print(False)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 1,2,3,5,8<br>6 | False | False | ✓ |
| ✓ | 3,5,9,45,42<br>42 | True | True | ✓ |
| ✓ | 52,45,89,43,11<br>11 | True | True | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

**Constraints**

1<= string length <= 200

**Sample Input 1**

experience
enc

**Sample Output 1**

xpri

**Answer:** (penalty regime: 0 %)

```
s1 = input()
s2 = input()
for c in s1:
    if c not in s2:
        print ( c, end="")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | experience<br>enc | xpri | xpri | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

**Example 1:**

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.
```

**Example 2:**

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums.
```

**Example 3:**

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
```

**For example:**

| Test | Result |
|---|---|
| `print(missingNumber([3,0,1]))` | 2 |
| `print(missingNumber([0,1]))` | 2 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def missingNumber(numbers):
    n=len(numbers)
    return n*(n+1)//2-sum(numbers)
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | `print(missingNumber([3,0,1]))` | 2 | 2 | ✓ |
| ✓ | `print(missingNumber([0,1]))` | 2 | 2 | ✓ |
| ✓ | `print(missingNumber([9,6,4,2,3,5,7,0,1]))` | 8 | 8 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

**For example:**

| Test | Result |
|---|---|
| `print(BalancedStrings('RLRRLLRLRL'))` | 4 |
| `print(BalancedStrings('RLLLLRRRLR'))` | 3 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s):
    c = 0
    b = 0
    for i in s:
        if i=='R':
            b+=1
        else:
            b -= 1
        if b == 0:
            c+=1
```

```
    return c

```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | print(BalancedStrings('RLRRLLRLRL')) | 4 | 4 | ✓ |
| ✓ | print(BalancedStrings('RLLLLRRRLR')) | 3 | 3 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

String should contain only the words are not palindrome.

**Sample Input 1**

Malayalam is my mother tongue

**Sample Output 1**

is my mother tongue

**Answer:** (penalty regime: 0 %)

```
s = input().split()
for w in s:
    x = w.lower()
    if x != x[::-1]:
        print(x,end=" ")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | Malayalam is my mother tongue | is my mother tongue | is my mother tongue | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.